

Ayman Hatem Zahran

AWS · Azure · GCP · VMware · Kubernetes — Cloud / DevOps / Platform / SRE / AI Freelancer · AWS Community Builder · AWS Certification SME · Kubestronaut · 11× AWS · 5× Azure · 2× GCP · 4× VMware · 5× Kubernetes · 3× HashiCorp · MBA · DBA

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SUMMARY

Cloud / DevOps / Platform / SRE / AI **Freelancer** with comprehensive cloud-and-container expertise across AWS, Azure, GCP, VMware, and CNCF. Proven record leading Kubernetes orchestration at scale, CI/CD automation, multi-cloud migration, and disaster-recovery design achieving 99.95% uptime SLAs. Build resilient, self-healing infrastructure under the "Everything as Code" philosophy — automate everything, eliminate waste, free engineering bandwidth for innovation. Active **AI/LLM practitioner**: production multi-provider LLM integrations (Claude · OpenAI · Gemini), MCP (Model Context Protocol) tool servers, agentic workflows, AI-assisted IaC, and AI-augmented CI/CD pipelines. Shipped AI SaaS apps: [merge.news](#) (AI-curated cloud/DevOps/AI brief) and outbox (AI-assisted social-media scheduler). AWS Community Builder, AWS Certification SME, and Kubestronaut.

EXPERIENCE

Tech Lead

Oct 2023 – Present

Virtasant

Austin, TX (Remote)

- Architected multi-account migration from public cloud to private cloud, improving resource allocation and IP-space utilisation across all production tiers.
- Led a team of 4–5 SREs managing critical CI/CD infrastructure with **1,000+ daily pipeline executions**; established IaC standards reducing deployment time by **75%** across **20+ Kubernetes clusters**.
- Designed DR/HA strategies across **2 regions** achieving **99.95% uptime SLA**; ran bi-weekly maintenance windows with zero customer-visible downtime.
- Automated compliance checking + vulnerability remediation cutting security findings by **90%**; standardized deployments across **25+ microservices** reducing failure rates by **60%**.
- Authored **30+ runbooks** and **20+ operational procedures** dropping MTTR by **40%**; managed **15+ P1 incidents** with zero SLA breaches.

DevOps Engineer

Oct 2022 – Oct 2023

Virtasant

Austin, TX (Remote)

- Managed and optimised **30+ Kubernetes clusters**; performed quarterly upgrades across **11 versions (1.21 → 1.32)**; automated updates for **100+ infrastructure components** cutting manual effort by **80%**.
- Migrated container networking across **10+ EKS clusters** improving performance by **30%**; upgraded CI/CD engines across **10+ environments** supporting **500+ daily builds**.
- Built monitoring stack with Prometheus / AlertManager / Grafana tracking **200+ custom metrics**; achieved **100% PSA compliance**.
- Managed bi-monthly secret rotation for **50+ service accounts** using Vault; rolled out GitOps across **12 clusters** enabling **20+ daily deployments**.
- Drove cost-optimisation programme achieving **35% reduction** through storage tiering, autoscaling, and right-sizing.

DevOps Lead

Feb 2023 – Jun 2023

Netways

Dubai, UAE

- Led DevOps transformation for Azure Cloud infrastructure across **5+ projects**: cut operational costs by **40%**, improved deployment frequency by **5×**.
- Built and optimised **50+ Azure DevOps pipelines** and **20+ repositories**; release cycle moved weekly → daily.
- Implemented IaC with Terraform across **5+ Azure environments**; reusable modules reduced provisioning time by **70%**; **25+ Helm charts** achieving **99.9% deployment success**.
- Mentored **5+ engineers** via bi-weekly workshops (+35% productivity); engaged **10+ stakeholder groups** with **95% CSAT**.

DevOps Consultant

Oct 2022 – Feb 2023

Syntheia

Sydney, AU (Remote)

- Delivered DevOps engagements for **3+ enterprise clients**; deployment cycles –60%, system reliability 99.95%.
- Deployed **5+ production EKS/AKS clusters** across AWS and Azure with FluxCD GitOps reducing manual intervention by **80%**.
- Authored **15+ Helm charts** reducing config errors by **75%**; produced **30+ technical guides**.
- Automated multi-cloud Kubernetes provisioning with Terraform — **12+ clusters** deployed in <30 min via reusable modules.
- Established CI/CD with integrated security scanning achieving **zero critical vulnerabilities** in production.

L3 DevOps Engineer

Apr 2022 – Oct 2022

Rackspace Technology

Cairo, Egypt

- Designed cloud architecture for **10+ microservices** on AWS reducing infrastructure cost by **45%**.
- Automated deployment pipelines with Jenkins / GitHub Enterprise / Ansible: release time 4h → 30min, **50+ deployments / week**.
- Migrated **12 monolithic applications** to microservices on EKS — scalability +70%.
- Implemented security best practices across **15+ AWS accounts** achieving **SOC2 compliance**.

L2 Professional Services Delivery Engineer

Dec 2021 – Apr 2022

Rackspace Technology

Alexandria, Egypt

- Delivered **25+ AWS solutions** for enterprise clients reducing deployment from weeks to hours.
- IaC with Terraform / CloudFormation / Runway / Serverless framework — **90% reduction** in manual config errors; architected across **20+ AWS services** optimising cost by **35%**.
- Migrated **4 on-premises workloads** to AWS reducing operational cost by **40%**.

Technical Support Engineer

Feb 2021 – Nov 2021

Dell Technologies

Cairo, Egypt

- Resolved **200+ monthly tickets** for VBlock/VxBlock with **95% first-call resolution**; SLAs exceeded by **20%**.
- Supported VMware (vSphere, NSX, vRealize), EMC storage (VMAX, VNX, Isilon), Cisco UCS / Nexus / MDS, Dell PowerEdge.
- Managed VPLEX clusters and RecoverPoint replication for **100+ TB data** at **99.9% availability** across **30+ deployments**.

Technical Support Engineer

Aug 2020 – Feb 2021

Vodafone

Cairo, Egypt

- Delivered 24/7 support for Telecom IN / Mediation infrastructure — **99.9% uptime** across **50+ production nodes**.
- Administered Ericsson IN (CCN, OCC, SDP, AIR, VS, EMM); commissioned **15+ new nodes** (deployment time –30%).
- Executed **20+ quarterly upgrades** supporting **40% traffic growth**; integrated SPLUNK monitoring across **12 nodes** (detection time –60%).

Full Stack Developer

May 2018 – Jul 2020

Echo Egypt

Cairo, Egypt

- Built and deployed **5+ desktop and web applications** end-to-end achieving **99% CSAT** and **35% faster delivery**.
- Enterprise solutions with C# / .NET / IIS / Microsoft AD / SSMS — **+40% operational efficiency**, **100K+ daily transactions**.

EDUCATION

Doctor of Business Administration (DBA)

Jun 2025 – May 2027

Swiss School of Business and Management Geneva (SSBM Geneva) — in progress

Dissertation: *From Human Freelancers to Autonomous AI Agents: A Framework for Assessing Efficiency, ROI, and Business Transformation in Freelance Platforms.*

- Research project: a framework for evaluating how freelance platforms adopt AI, and what that does to **operational efficiency, ROI, and business-model transformation**.
- Studies **hybrid platforms** where AI does not simply replace human freelancers — examples include **Upwork, Fiverr, Toptal, and 99designs**.
- Combines a conceptual framework linking AI adoption to platform performance, quantitative research (surveys of platform-side professionals), and validation of that framework.
- Core argument: this market needs a **sector-specific** way to measure AI's effect; generic digital-transformation models are too broad.

Master of Business Administration (MBA), Digital Transformation

Oct 2023 – Dec 2024

Zigurat Innovation & Technology Business School · Universitat de Barcelona

Thesis: *Revolutionizing IT Infrastructure: Innovating Cloud and DevOps Solutions Through Polyglot Programming Paradigms.*

- FMT / capstone vehicle was **EchoOps**: a designed Cloud / DevOps startup (not an employer) around **Everything as Code** and **polyglot programming** (right language per layer), plus automation, GitOps, and multi-cloud delivery.
- Combined business strategy, an operating model, primary research, and a financial model. Financials are **thesis projections**, not live company results.
- Core argument: polyglot + Everything as Code is a practical **infrastructure-transformation strategy**, not only a tooling preference.

Bachelor's Degree, Communication and Computer Engineering

2012 – 2018

Cairo University, Faculty of Engineering — 5-year integrated programme

- Core curriculum across **signals & systems, digital electronics, computer architecture, operating systems, networks, embedded systems, and software engineering**.
- Mastered **C# · Java · Python · MATLAB · Assembly · SQL · VHDL** through **20+ hands-on projects** spanning OS kernels, network protocol stacks, hardware-description synthesis, and full-stack web applications.
- Completed **270+ hours of industrial training** with leading tech companies in Cairo; rotations in network operations, software QA, and embedded-systems R&D.
- Senior capstone: end-to-end system design integrating embedded hardware, a custom protocol layer, and a desktop control application — demonstrated cross-layer debugging and ownership.

Scholarship · AWS Practical Data Scientist Academy

Oct 2021 – Mar 2022

Ministry of Communications and Information Technology (MCIT), Cairo, Egypt — government-sponsored intensive scholarship

- Selective scholarship covering **practical applied data science** on AWS: SageMaker, Glue, Athena, Comprehend, Kinesis, QuickSight.
- End-to-end ML lifecycle: feature engineering, model training and tuning, deployment as inference endpoints, monitoring and re-training automation.
- Capstone projects built on real datasets; outputs evaluated on accuracy, cost-per-inference, and operational robustness.

Scholarship · Datacenter Infrastructure & Virtualization

Feb – Apr 2020

National Telecommunication Institute (NTI), Cairo, Egypt — government-sponsored intensive scholarship

- Hands-on programme on enterprise datacenter operations: **VMware vSphere / vCenter / NSX / vSAN**, Cisco UCS compute, Nexus / MDS networking, Dell EMC storage (VMAX / VNX / Isilon).
- Foundation that later underpinned Dell Technologies VBlock / VxBlock support work and the L3 cloud architect role at Rackspace.

CERTIFICATIONS (40+)

AWS — 11

Solutions Architect Professional · DevOps Engineer Professional · Solutions Architect Associate · Developer Associate · SysOps Administrator Associate · Cloud Practitioner · Security Specialty · Machine Learning Specialty · Data Analytics Specialty · Database Specialty · AI Practitioner

Kubestronaut · 5× Kubernetes + 2× CNCF

CKA · CKAD · CKS · KCNA · KCSA · Argo Project Associate · Prometheus Associate

Microsoft Azure — 5

Administrator Associate (AZ-104) · Fundamentals (AZ-900) · AI Fundamentals (AI-900) · Data Fundamentals (DP-900) · DevOps Solutions Expert (AZ-400)

Google Cloud — 2

Cloud Digital Leader · Generative AI Leader

VMware — 4

VCP-DCV (2020 & 2021) · VCP-NV (NSX-V & NSX-T) · Digital Business Transformation Associate · vSphere 6.7 Foundations

HashiCorp — 3

Terraform Associate · Vault Associate · Consul Associate

IBM — 5

Containers & Kubernetes Essentials · Docker Essentials · IBM Cloud Essentials · Applied Data Science with Python (Levels 1 & 2) · Deep Learning with TensorFlow / Accelerated DL with GPU

Other

Cisco CCNA · Dell Technologies Associate (ISM v4.0) · Rackspace MultiCloud Practitioner · Triplebyte Certified · Kaggle — 10+ ML/DL/Data micro-certificates

TECHNICAL SKILLS

AI & LLM engineering. Anthropic Claude (Opus / Sonnet / Haiku) · OpenAI (GPT-4o / GPT-5) · Google Gemini · multi-provider LLM routing with per-task model selection · MCP (Model Context Protocol) tool servers · agentic workflows + parallel sub-agent orchestration · retrieval-augmented generation · function calling · prompt engineering for IaC · AI-assisted code review · Amazon Bedrock · AWS SageMaker · Google Vertex AI · Hugging Face · LangChain · Claude Code CLI / Claude Agent SDK

Cloud platforms. AWS (Compute / Network / Storage / Database / Security / ML / AI) · Microsoft Azure · Google Cloud Platform · Firebase / App Hosting · VMware vSphere

IaC & provisioning. Terraform · Pulumi · Terragrunt · env0 · Ansible · Packer · Vagrant · CDK · CDKTF · CDK8s · Projen · CloudFormation · Serverless (EB / SAM / Amplify) · Runway

Containers & orchestration. Docker · Kubernetes · Helm · EKS · ECS · EKSCTL · AKS · GKE · Azure ACR · AWS ECR

CI/CD & GitOps. Jenkins · GitHub Actions · GitLab CI · Azure Pipelines / DevOps / Boards / Repos · Tekton · Spinnaker · ArgoCD · FluxCD · JFrog · BitBucket

Observability. Prometheus · AlertManager · Grafana · OpenTelemetry · ELK · CloudWatch · SPLUNK

Security & compliance. HashiCorp Vault · AWS IAM / KMS / Security Groups · OPA · PSA · SOC2 · GDPR · vulnerability remediation · secret rotation

Programming. PowerShell · Bash · PowerCLI · Python · TypeScript · JavaScript · C# · C / C++ · VB.NET · Groovy · HCL · YAML · JSON · HTML / CSS · PHP · SQL (MSSQL / MySQL)

Virtualization & datacenter. VMware vSphere / vCenter · NSX-V / NSX-T · vSAN · Cisco UCS B/C-Series · Nexus / MDS Switches · Dell V/VxBlock · PowerEdge · EMC VMAX / VNX / Isilon / XIO · Unity · VPLEX / RecoverPoint

Telecom. Ericsson IN (CCN, OCC, SDP, AIR, VS, EMM) · GGSNs · MSCs · Mediation Nodes · SPLUNK telemetry

Operating systems & networking. Linux (RHEL / Ubuntu / Amazon Linux) · Windows Server 2016+ · Microsoft AD · IIS · SSMS · DNS · DHCP · TCP/IP · VPN · Firewall

PERSONAL AI / SAAS PROJECTS

merge.news — AI-curated Cloud / DevOps / AI brief

2025 — Present

- Cloud Functions (Node 22) generate articles every 15 minutes via **Gemini + Google Search grounding**; multi-provider LLM router (Claude · OpenAI · Gemini) for chatbot, tweet compression, and article generation.
- Next.js 15 App Router web client + Expo Router mobile app; Firestore + Firebase Auth + App Hosting; 3-tier filter (Category → Subcategory → Hashtag), sort controls on every page, public OG/Twitter Card meta.
- Lemon Squeezy subscription billing (\$5/mo Pro tier); admin-managed schedule + slot-based posting to X (OAuth 1.0a) + LinkedIn Community Management API.

outbox — AI-assisted social-media scheduler

2025 – Present

- tRPC 11 + Vite React + Express server packaged for Cloud Functions; Expo Router mobile mirror with shared auth contract.
- Multiple AI providers (Grok · OpenAI · Claude · Gemini) for AI-generated post drafts; per-platform compose modes (X, LinkedIn share, LinkedIn comment, quote, reply).
- Scheduled posting via Cloud Functions + slot-based scheduler running every 60s; deep-link /compose contract so other apps (merge.news admin) can push drafts in.

COMMUNITY & VOLUNTEER INVOLVEMENT

AWS Authorized Instructor (AAI)

Jun 2023 – Present

- Authorized AWS instructor delivering **5+ official training courses** on cloud architecture, DevOps, serverless.
- Completed **40+ hours** of certification covering **20+ AWS services** (Lambda, ECS/EKS, CloudFormation, automation).

AWS Certification SME / Exam Writer

Apr 2022 – Present

- Reviewed and validated **10+ AWS certification exam questions** for Solutions Architect and DevOps Engineer paths.
- Contributed to **5 exam development workshops** across **3 certification tracks**.

AWS Community Builder — Cloud / DevOps

Mar 2022 – Present

- Selected for the Cloud / DevOps track among **500+ global applicants**.
- Engaged **1,000+ community members**; contributed **5+ technical articles** on serverless, container orchestration, IaC.

LANGUAGES

Arabic. Native or bilingual proficiency · **English.** Professional working proficiency · **French.** Elementary

`git commit -m "Everything as a Code"` · Let's automate the future together — BYOD